NANYANG TECHNOLOGICAL UNIVERSITY
School of Social Sciences
HE3002 Macroeconomics III

PROBLEM SET 1
The Open Economy: Openness in Goods and Financial Markets

1. Using the information in Lecture 1, label each of the following statements true, false or uncertain. Explain briefly.

   (a) Although the export ratio can be larger than one, as it is in Singapore, the same cannot be true of the ratio of imports to GDP.

   (b) Uncovered interest parity implies that interest rates must be the same across countries.

   (c) The nominal exchange rate and the real exchange rate always move in the same direction.

   (d) If the dollar is expected to appreciate against the yen, uncovered interest parity implies that the US nominal interest rate must be greater than the Japanese nominal interest rate.

2. Show that by approximation,

$$(1+i_t) = \frac{(1+i_t^*)}{[1+(E_{t+1}^e - E_t)/E_t]}$$

   can be expressed as:

$$i_t = i_t^* - \frac{E_{t+1}^e - E_t}{E_t}$$

3. Each of the government of Brazil and Turkey has issued bonds in Brazillian real (BRL) and Turkish lira (TRY), respectively. Assume that both government securities are one-year bonds, i.e. paying the face value of the bond one year from now. Suppose that the exchange rate, E, stands at 1 Brazillian real per 0.79 Turkish lira. The following table gives face values and prices of both bonds:

	Face Value	Price
Brazil	BRL10,000	BRL9,630
Turkey	TRY10,000	TRY9,450

   (a) Compute the nominal interest rate on each of the bonds.

   (b) Compute the expected exchange rate next year consistent with uncovered interest parity. If you expect the real to depreciate relative to lira, which bond should you buy.

   (c) Assume that you are a Brazilian investor. You exchange the reals for liras and purchase the Turkish bond. One year from now, it turns out that the exchange rate, E, is 0.75 i.e. 0.75 liras buy one real. What is your realized rate of return in reals compared to the realized rate of return you would have made had you held the Brazilian bond?

   (d) Are the differences in rates of return in part (d) consistent with the uncovered interest parity condition? Explain your answer.